American Journal of Epidemiology, 2026, 00(00), 1–11
https://doi.org/10.1093/aje/kwag204
Published: August 24, 2026
Novel Methodology

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Improving dietary data quality in nutrition studies: development and validation of an explainable, reproducible, open machine learning-assisted framework for outlier detection in 24-hour recalls

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Abstract

Twenty-four-hour dietary recalls (24HRs) are widely used in nutritional epidemiology, but repeated administration is required to capture within-person variability, introducing temporal dependence and challenges for data cleaning. Existing outlier detection approaches are largely static, manual, and based on adult-centered thresholds, limiting reproducibility and scalability. We developed and validated an explainable, reproducible machine learning (ML)-assisted framework to detect and evaluate implausible intake values in cross-sectional and longitudinal 24HRs and to generate artificial intelligence (AI)-ready dietary datasets. The open pipeline integrates automated statistical detection using digitized National Cancer Institute 5th-95th percentile intake ranges, longitudinal IQR-based flagging, and ML-based rule explainability for outlier detection. Validation was conducted using the MIcrobiota, GROWth and Diet study in Ontario, Canada (126 children aged 8-12 years), and data from the 2015 Canadian Community Health Survey-Nutrition, a nationally representative sample of Canadian children and adults with repeated 24HRs ($n = 15\,216$). Across energy intake and 7 nutrients, 75 recalls containing at least 1 potentially implausible value were identified. In longitudinal data, the framework achieved >77% sensitivity, 97% specificity, and 88% precision, outperforming the static method. This explainable data-driven framework improves data quality, transparency, and reproducibility, supporting scalable epidemiologic analyses using AI-ready dietary data.

Key words data-driven outlier detection, 24-hour dietary recall, longitudinal diet data, child and adult nutrition, AI-ready diet data

Received: April 29, 2026. **Accepted:** August 19, 2026

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Introduction

Twenty-four-hour dietary recalls (24HRs) are widely used to assess dietary intake.[1] Although they capture detailed short-term consumption, they are limited by within-person variability and may not reflect usual intake. Multiple recalls collected at different time points are essential to study usual dietary intakes, allowing for intra-individual variation and reducing measurement error.[2] The widespread use of single or multiple 24HRs is facilitated by the freely available web-based Automated Self-Administered 24-hour (ASA24®) Dietary Assessment Tool developed by the National Cancer Institute (NCI), which can be completed independently and inexpensively.[3] As of November 2024, ASA24® has been used in over 1000 peer-reviewed publications.[4]

Despite their strengths, 24HRs introduce analytical challenges, including their sensitivity to within-individual day-to-day dietary variation and noise from episodically consumed foods,[5] which can complicate data analysis and interpretation.[6] This limitation may be addressed by multiple collections from the same individual at various time points.[7] Although numbers vary across study designs,[8,9] between 3 and 7 collections are typically conducted to adequately estimate energy intake.[10-12] However, while necessary, the repeated measures can pose data preprocessing and outlier detection challenges.[13]

Outliers in dietary data may arise from data entry errors, reporting issues, or inaccuracies in nutrient databases. However, extreme values may also reflect true dietary behaviors and provide meaningful insight, such as individual or population-level eating patterns.[5] Current guidance from the NCI emphasizes careful review of such observations, as arbitrary exclusion may lead to loss of important information. Existing approaches for outlier detection include visual tools (eg, boxplots, scatterplots)[14] and empirical thresholds,[15] but these methods have limitations. Thresholds are often derived from specific populations (eg, the National Health and Nutrition Examination Survey-NHANES), limiting generalizability, and due to lack of digitization and automation, they are not easily integrated into automated pipelines, reducing reproducibility.

Importantly, existing approaches treat each recall as an independent observation, ignoring relationships between dietary intakes over time.[16] Repeated dietary recalls introduce temporal dependencies and require preprocessing methods specifically designed for time-series data.[17] A recent study from our group showed that the inclusion of temporal associations in data-cleaning algorithms increases outlier detection accuracy.[18] Yet, this approach has not been applied to dietary data.

To address this gap, this study developed a reproducible software pipeline for preprocessing and cleaning dietary recall data, including estimation of outlier thresholds for both adults and children. Rather than discarding flagged observations, the approach prioritizes their evaluation to minimize unnecessary data loss ("overtossing"). The proposed framework combines transparent statistical outlier detection with explainable machine learning (ML) to support the generation of AI-ready dietary datasets. Here, AI-ready refers to quality-controlled dietary datasets with standardized preprocessing, documented outlier flags, and reproducible data-cleaning decisions that support downstream statistical and ML applications.[19] Canadian data are used as a case study to demonstrate adaptation to a specific population, with a framework designed to be transferable to other settings with appropriate calibration.

Methods

Datasets

Children dataset, the MIcrobiota, GROWth and Diet (MiGrowD) study

We included 126 children of both sexes aged 8-12 years inclusive enrolled in MIcrobiota, GROWth and Diet (MiGrowD).[20] MiGrowD is embedded in The Applied Research Group for Kids! (TARGet Kids!), a practice-based research network in Ontario, Canada.[21] Children completed the recalls using the ASA24®-Canada NCI tool. We used all available 24HRs from MiGrowD study participants collected during their longitudinal participation in the TARGet Kids! cohort. Participant characteristics and number of recalls are summarized in Table S1.

Reference population dataset, the 2015 Canadian Community Health Survey (CCHS)-Nutrition

This is a population-based national survey that included 20 487 participants aged 1 to >95 years.[22] Diet data were collected through interviewer-administered 24HRs completed using the same multiple-pass method that informed ASA24®.[23] A subsample of 7608 respondents completed a second 24HR, 3 to 10 days later. Here, we used both the entire 2015 Canadian Community Health Survey (CCHS)-Nutrition dataset and a subsample including only children aged 8-12 years (the MiGrowD participants' age range).

The MiGrowD dataset provides longitudinal, individual-level dietary intake data, allowing for analysis of within-person variability. The 2015 CCHS-Nutrition dataset serves as a population-representative reference, enabling comparison between MiGrowD participants and a broader Canadian population of children; it also allowed us to estimate reference values for the general population. Both datasets used validated dietary assessment tools, ensuring consistency in methodology and reducing measurement error. In addition, combining the datasets creates a diverse sample that includes both children and adults, allowing us to generate more generalizable estimates. An overview of the datasets used in this study, including their sample sizes, age ranges, recall structure, and analytic role, is provided in Table 1.

Systematic pipeline for the detection, understanding, and evaluation of outliers

The new software pipeline has 3 components (Figure 1): (1) outlier detection, for both cross-sectional and longitudinal data; (2) outlier exploration, where an explainable ML method is applied to understand the root cause of each outlier, the relationship between different nutrient outliers, and the relationship between outliers and dietary habits; (3) evaluation of outlier detection methods, including data preprocessing, injection of artificial outliers (AO), execution and comparison of outlier detection methods, and their evaluation. Overall, the proposed framework follows a hybrid statistical-ML approach. Primary outlier detection relies on transparent statistical methods, whereas explainable ML is optionally used to derive interpretable dietary decision rules and identify patterns associated with flagged observations.

1) Outlier detection

The initial phase of the outlier detection component involved digitizing the NCI/ASA24® recommendations,[14] using automatically executed R code (Appendix S1). These thresholds were originally derived

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Table 1 Overview of datasets used for reference-value derivation, outlier detection, and validation.

Dataset	Population	Age range	Participants/respondents	24HRs	Data structure	Role in analysis
MiGrowD	Children enrolled in the MIGrowD study in Ontario, Canada	8-12 years	126 participants initially; 122 participants after preprocessing	251 recalls initially; 226 recalls after removing incomplete and duplicate recalls	Longitudinal/repeated recalls; 1-4 recalls per participant	Primary study dataset for outlier detection, longitudinal evaluation, and CART-based outlier exploration
2015 CCHS-Nutrition (all ages)	Nationally representative Canadian children and adults with repeated recalls	1 to >95 years	7608 respondents with 2 recalls	15216 recalls	Repeated 24HRs recalls; 2 recalls per respondent	Reference dataset for deriving age- and sex-specific intake thresholds and external validation
2015 CCHS-Nutrition (child subset)	Age-matched Canadian children	8-12 years	1182 respondents with 2 recalls	2364 recalls	Repeated 24HRs recalls; 2 recalls per respondent	Age-matched validation dataset for comparison with MiGrowD children

Abbreviations: 24HRs, 24-hour dietary recall; CART, classification and regression tree; CCHS, Canadian Community Health Survey; MiGrowD, MIcrobiota, GROWth and Diet.
MiGrowD sample sizes are shown both before and after preprocessing. Preprocessing removed incomplete recalls ($n = 18$) and duplicate recalls ($n = 7$), resulting in 226 cleaned recalls from 122 participants.

from NHANES data for triaging dietary data and were used in this study as a secondary reference rather than rigid exclusion cut points, as extreme values in short-term instruments such as 24HRs may reflect true intakes. Because the ASA24® documentation does not specify the exact NHANES cycles used and includes thresholds for a limited number of nutrients, we implemented age- and sex-specific reference values from the 2015 CCHS-Nutrition as our primary standard.

The automated code streamlined the implementation of the cleaning methods, improving efficiency and standardizing the outlier detection processes independently of study content and design. Subsequently, we compared 3 outlier detection methods using energy intake (kcal) and 7 nutrients, for which NCI includes thresholds in the guidance (carbohydrates, fiber, sugars, total fat, proteins, sodium, and vitamin C). The first method, based on the ASA24®/NCI 5th-95th percentile guidance, served as the standard guidance-based method. Specifically, a value was considered an outlier if it was below the 5th percentile or above the 95th percentile for a given variable. These cutoffs were selected to identify observations outside the central 90% of the age- and sex-specific reference distribution provided by the ASA24®/NCI data-cleaning guidance, allowing them to be flagged for further evaluation.[14] Age- and sex-specific reference values were automatically derived for both children and adults from the 2015 CCHS-Nutrition dataset.[22] Population-specific reference values should be used when applying the pipeline in other contexts.

The second method adapted the biologically implausible value (BIV) detection method previously described elsewhere.[18,24] This method flagged a value as an outlier if it exceeded the 75th percentile plus twice the IQR of the reference distribution. We replaced the standard deviation in the original BIV method with IQR because, compared to standard deviation, it is less influenced by extreme values and does not rely on the assumption of normality, making it more appropriate for dietary intake data. The multiplier of 2 retains BIV's intended stringency.[25,26]

The third method extended this approach to account for longitudinal patterns. Flagged values from the second method were compared with values from preceding and subsequent ASA24® recalls for the same respondent. If no other values exceeded the difference between the flagged value and twice the IQR, the flagged value was reclassified as part of a series of recalls potentially warranting contextual review by an expert. This approach does not assume progressive change but rather repeated occurrence, making it suitable for dietary intake data.

2) Outlier exploration

We examined flagged recalls for distinct nutrient combinations to distinguish data-entry errors from plausible extremes. This also allowed us to distinguish them from actual outliers. For example, a combination of high calories, carbohydrates, and sugars may indicate a recall high in sweets and desserts rather than a random error. We trained a decision tree (DT) classifier using the classification and regression trees (CARTs) algorithm[27] and the rpart package in R,[28] aiming to extract interpretable decision rules for classifying observations as outliers or non-outliers. Briefly, CART was selected because it produces simple threshold-based rules that can also be interpreted by non-ML experts. Rather than providing only a predicted probability, DTs can show which food-group intake thresholds helped classify a recall as an outlier.[29] The model used Gini impurity as the splitting criterion to decide which food-group variable, and which intake threshold, best separated recalls into groups that were more or less likely to contain flagged outliers. Details of model selection and evaluation

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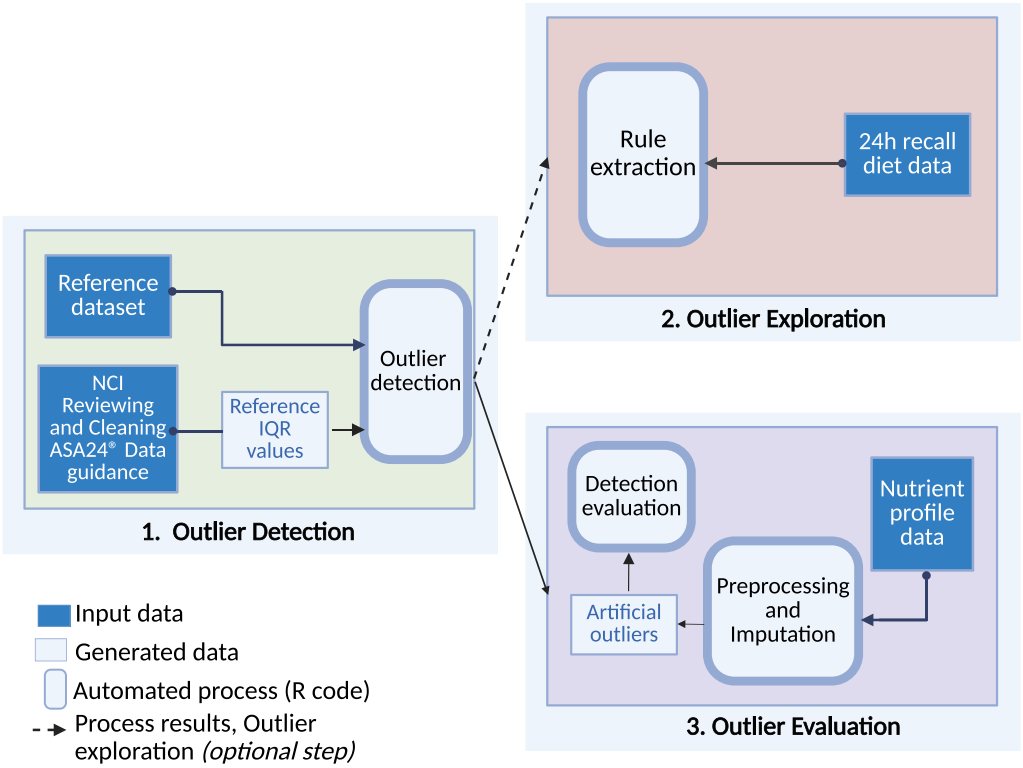


Figure 1 A systematic software pipeline for detecting, understanding, and evaluating outliers in ASA24® 24-hour dietary recalls. The pipeline consists of 3 components: (1) Outlier detection (dietary intake values from the reference dataset—here, the 2015 CCHS-Nutrition—and the NCI guidelines are used to compute reference IQR values. These are then used to detect potential outliers). (2) Outlier exploration (dietary patterns contributing to outliers are identified through interpretable decision rules, utilizing ASA24® 24-hour dietary recall data and explainable ML). (3) Outlier evaluation (artificially generated outliers are introduced for validation. The detected outliers undergo preprocessing and imputation, followed by comparison with nutrient profile data to assess the pipeline's performance). The reference dataset used to extract reference values for this study was the 2015 CCHS-Nutrition dataset; future studies can use the pipeline with different reference datasets. Created with BioRender. Massara, P. (2026). https://BioRender.com/yv71qv4. ASA24®, Automated Self-Administered 24-hour; CCHS, Canadian Community Health Survey; ML, machine learning; NCI, National Cancer Institute.

are provided in Appendix S2. The CART model was applied as an explainable ML component for exploratory rule extraction. As predicates for the rules, we used the ASA24® foods categories, including major food groups such as "fruit," "vegetables," "grains," "proteins," "dairy," "oils," and "added sugars." While this approach provides some degree of correlation, not of causality, it provides insight into common food combinations that may lead to outliers.

3) Outlier evaluation

The third component of the pipeline aimed to evaluate the aforementioned outlier detection methods. While in this study we focused on these 3 outlier detection methods, this component was designed to evaluate any other outlier detection method systematically. Several performance metrics were calculated after applying the detection method, including sensitivity and specificity, as well as precision and recall. By comparing multiple methods, commonly detected outliers can be assessed.

For unbiased and reproducible testing, datasets were cleaned before injecting AO. The data-cleaning workflow is described in detail in Appendix S3. Briefly, this first included several preprocessing steps for identifying structural errors, including broken or duplicate entries and incomplete textual descriptions (generally representing data-entry or program-related issues that cannot be meaningfully interpreted). Next, we focused on distribution-based outliers in alcohol and caffeine intake, and outliers in specific food groups such as snacks or beverages.[14] While both steps reduce data noise, the former addresses non-nutritive data quality problems, while the latter includes decisions about extreme but potentially valid intakes. Additionally, the tool conducted a missingness analysis which recommended the removal of columns or rows with a high percentage of missing values (>10% of values). Naturally occurring extreme nutrient values were flagged but were not treated as validation ground truth because their true status as errors or plausible extreme intakes could not be known with certainty.

Using the NCI guidance, values outside the 5th-95th percentile of the primary 2015 CCHS-Nutrition reference distribution were flagged as "not available" (NA) and replaced with imputed values. Winsorization, which replaces extreme observations with predefined boundary values (eg, the 5th or 95th percentile), was not applied as it could effectively hide outlier values, unusual dietary patterns, data-entry errors, or genuinely extreme intake values.[30,31] Instead, these observations were retained to preserve the integrity of the subsequent outlier detection process. Moreover, to ensure that the imputed values were representative of the real data, we compared 3 imputation methods: weighted predictive mean matching, random forest, and lasso linear regression, as implemented in the mice R package.[32] Each method was applied with 15 multiple imputations to minimize Monte Carlo error and ensure stable parameter estimates.[33] The average across

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the 15 imputations was selected as the final value for each missing observation. For each of the 7 nutrients, we compared the mean value, the standard deviation, the IQR, and the difference in IQR between the clean dataset (without outliers) and the imputed dataset. These metrics were calculated for 3 datasets (the original, cleaned, and imputed) for each of the 15 multiple imputations. Statistical differences between methods and datasets were assessed using analysis of variance (ANOVA), while the performance of imputation methods relative to the cleaned data was evaluated using a *t*-test.

The outlier detection methods were tested in all datasets using AO to better control the validation process. A subset of the dataset, including 20% of the original data, was randomly selected to be replaced with outlier values. The outlier injection process was designed based on Tukey's "5-number summary"[26] using the percentiles of the data distribution (5th, 25th, median, 75th, 95th). For every random value replaced by an outlier, we checked the sex and age of the respondent to determine the reference value of the IQR that would be used to define the outlier value. Performance metrics (sensitivity, specificity, precision, recall) were calculated for all methods. Two types of outliers were introduced:

1) *Single outlier cases occurring in an individual (static outliers).* Two subtypes were injected. The first was "moderate-to-extreme," where the original value was increased by adding a random multiple of the corresponding reference IQR value, in the range of 2 or 3 times the IQR (+[2-3] × IQR). Depending on the original value, the outlier may not be as extreme, but it is still expected to be an abnormal measurement, especially in the context of multiple recalls for an individual. The second subtype was "extreme," where the original value was selected randomly between the 75th and 95th percentiles of the reference population plus 3 times the reference IQR (>95th percentile +3 × IQR), as per the typical definition of an outlier in the 5-number summary.
2) *Multiple outlier cases occurring in an individual (longitudinal outliers).* If more than 2 24HRs by the same respondent and the same nutrient had flagged outliers, they were considered a pattern and were treated as non-outliers not to be removed. Thus, a modified longitudinal "ground truth" (the set of injected AO) was created to validate the outlier detection methods. Two outlier subtypes were injected as described previously. The temporal gap between recalls was not used as an exclusion criterion since we focused on the recurrence of flagged values for the same nutrient and the same respondent across all available 24HRs.

Validation

For every detection method to be validated and for every nutrient, a confusion matrix was constructed between the injected outliers (*reference*) and the detected outliers (*result*). From the number of true positives, true negatives, false positives and false negatives, we calculated the sensitivity, or true positive rate; specificity, or true negative rate, and balanced accuracy, which is the arithmetic mean of sensitivity and specificity. The formulas for calculating the performance metrics are described in Appendix S4. We also calculated common true positives, that is, injected outliers that were correctly identified by more than 1 method. This allowed us to assess the extent to which different methods identified the same true outlier observations. We evaluated 3 detection approaches: the digitized NCI guidance-based method, the longitudinal method with flagging, and the longitudinal method without flagging. We used 3 validation datasets: the MiGrowD analytic dataset, the full 2015 CCHS-Nutrition repeated-recall sample including participants of all ages, and an age-matched 2015 CCHS-Nutrition subset restricted to children aged 8-12 years. The 2015 CCHS-Nutrition dataset is curated; therefore, no preprocessing or data cleaning was applied. For the CCHS analyses, we retained only respondents who completed 2 24HRs. The rest of the process was performed as for the MiGrowD dataset, including outlier injection and detection with the 2 and 3 aforementioned methods, respectively. The same quality metrics were collected.

Results

Recall data

The MiGrowD dataset initially included 251 recalls from 126 participants. After removing incomplete (n = 18) and duplicate (n = 7) entries, the final analytic sample consisted of 226 recalls from 122 participants. Of these, 35 completed 1 recall, 72 completed 2, 13 completed 3, and 2 completed 4 (Table S1). All diet data from 2015 CCHS-Nutrition were included since they had already been curated.[22]

Naturally occurring outliers

After applying the NCI 5th-95th outlier detection method, we detected a total of 75 recalls with at least 1 naturally occurring outlier among the 7 nutrients or energy intake, with 3 respondents having more than 1 such recall. These counts reflect existing outliers detected prior to any AO injection. There were 17 outliers in energy intake, 24 in carbohydrates, 18 in fiber, 24 in sugars, 20 in total fat, 24 in proteins, 16 in sodium, and 10 in vitamin C. When the IQR outlier detection method was used, outliers were found only in vitamin C. Therefore, we consider only those found with 5th-95th for the remainder of the analysis.

Rule extraction

To explore outlier profiles, we used the ASA24® food groups as independent variables. DT leaf nodes contain the outcome of the classifier: "Y" for outlier, "N" for regular non-outlier value. Specifically, 57% of the outliers (with 73% probability) were linked to red/orange vegetable (other than tomato) consumption below 0.23 portions, fruits (citrus, melon, and berries) below 0.03 portions, milk below 1.9 portions, and starchy vegetable (potato) consumption below 0.45 portions. Another 9% of the outliers (with 86% probability) were explained by red/orange vegetable consumption below 0.23 portions, but fruit consumption above or equal to 0.03 portions and poultry consumption below 5 portions, along with egg consumption below 0.69 portions. Finally, 12% of the outliers (with 90% probability) were characterized by red/orange vegetable consumption below 0.23 portions, fruit consumption above or equal to 0.03 portions, poultry consumption above or equal to 5 portions, and egg consumption above or equal to 0.69 portions. These outlier profiles correspond to specific branches in the tree, with terminal nodes showing the proportion of entries following each path. Because the purpose of the CART analysis was exploratory rule extraction rather than predictive optimization, model performance was interpreted descriptively. The model showed limited discrimination, with an area under the receiver operating characteristic

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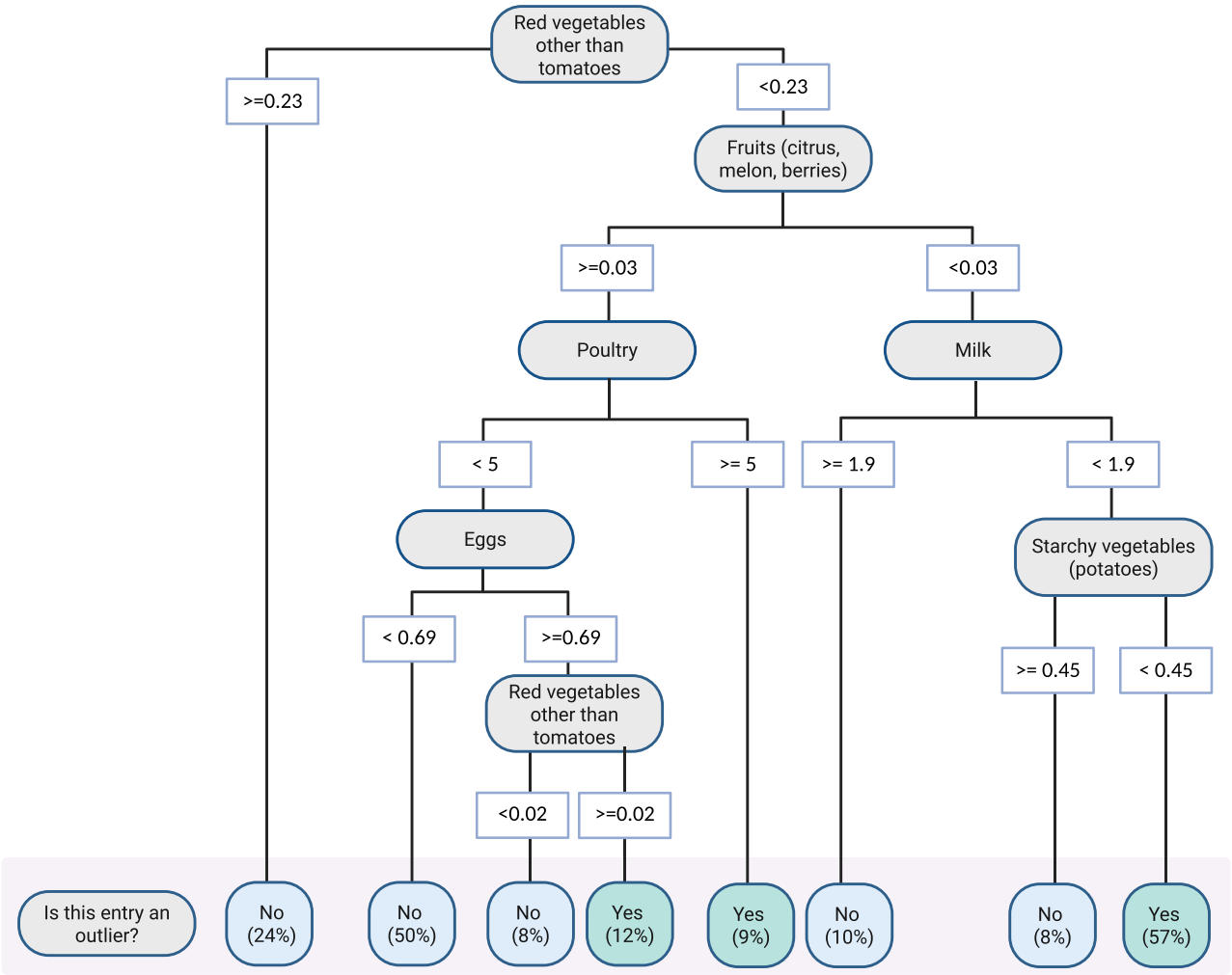


Figure 2 Decision tree criteria (rules) that show which food groups determine an outlier and their probabilities. This DT illustrates the classification of dietary records as outliers or non-outliers based on food group consumption, as captured by ASA24®. The model splits the data using thresholds for specific food categories (eg, poultry, milk, and starchy vegetables), sequentially refining classifications. The terminal nodes indicate whether an entry is an outlier (Y) or not (N), along with the percentage of records in each category. The model provides an interpretable approach to understand which food groups, and in which amounts, contribute to outlier classification. For example, green vegetables and red meat were not among the top predictors of outliers, suggesting that dietary variation in these categories may be more evenly distributed across individuals in our dataset. Created with BioRender. Massara, P. (2026). https://BioRender.com/c0ybudf. ASA24®, Automated Self-Administered 24-hour; DT, decision tree.

curve (Receiver Operating Characteristic–Area Under the Curve) of 0.62, and high sensitivity of 0.80. This suggests that CART was useful for identifying interpretable dietary patterns associated with flagged observations. Results are summarized in Figure 2.

Outlier injection and imputation in MiGrowD

After setting the 75 naturally occurring outliers into missing, we examined the effects of imputation. ANOVA showed that the 3 imputation methods produced different results ($P < .05$) in terms of the means of the 7 nutrients. Comparing the means of the imputed values against the means of the cleaned data (without outliers) with a *t*-test, the Random Forest imputation was found to produce the most consistent results compared to the cleaned data. Therefore, the rest of the analysis used the imputed data produced by this method.

Outliers were injected separately per nutrient. For each nutrient, we generated 45 outliers representing 20% of the original data as described previously, 22 of Type A (moderate-to-extreme) and 23 of Type B (extreme). Table 2 shows the impact of the outlier values on the mean, standard deviation, and IQR of each nutrient compared to the original imputed data.

Performance of outlier detection methods in MiGrowD

Table 3 shows the performance of the 3 detection methods evaluated in the MiGrowD dataset using the ground truth adapted for longitudinal recalls. In MiGrowD, the longitudinal method with flagging achieved high average sensitivity, specificity, precision, and balanced accuracy of 86.0%, 98.6%, 92.0%, and 92.1%, respectively, across nutrients. The static method showed similar average sensitivity and balanced accuracy of 84.6% and 91.5%, respectively, but lower specificity and precision of 96.9% and 83.6%. In contrast, the longitudinal method without flagging achieved the highest average sensitivity of 99.6%, but substantially lower specificity and precision of 55.5% and 29.2%, indicating a tendency toward over-detection.

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Table 2 Comparison of mean, standard deviation, and IQR before and after outlier injection across all nutrients.

	Before outlier injection			After outlier injection		
	Mean	**SD**	**IQR**	**Mean**	**SD**	**IQR**
Energy intake (kcal)	1790.1	523.9	777.9	2341.9	1299.6	1159.5
Carbohydrates	222.4	68.2	89.6	301.9	185.3	142.4
Fiber	15.5	5.7	8.9	21.3	13.4	12.1
Proteins	74.4	24.7	36.2	98.0	57.7	54.7
Sugars	92.5	36.4	50.5	138.9	108.1	77.5
Total fat	68.0	25.7	33.6	94.4	62.2	53.6
Vitamin C	86.5	66.3	91.4	167.0	188.7	134.9
Sodium	2706.6	930.8	1313.1	3696.6	2294.7	1999.9

Table S2 shows results using the original, non-adapted ground truth. With the adapted longitudinal ground truth, specificity and precision decreased across methods, but the longitudinal method with flagging maintained the best overall balance.

Validation in the 2015 CCHS-Nutrition datasets

Of the 20 487 respondents in the 2015 CCHS-Nutrition dataset, we first considered participants with 2 24HRs, resulting in 7608 respondents and 15 216 recalls. Artificial outliers were injected in 3043 recalls (~20% of the total dataset). The impact of the outlier injection in the 2015 CCHS-Nutrition data is shown in Table S3. To evaluate performance in a population comparable to MiGrowD, we also created an age-matched subset (children aged 8-12 years) including 1182 respondents with 2 recalls each. Outliers were injected in 473 recalls (~20% of the total dataset). Table S4 shows the impact of the outlier injection in this 2015 CCHS-Nutrition data subset. This procedure is summarized in Figure S1.

Performance results for the 2015 CCHS-Nutrition validation analyses are summarized in Table 4 for the full repeated-recall sample and Table 5 for the age-matched child subset. Sensitivity ranged from 84.3% to 99.4% in the full dataset and from 85.6% to 99.6% in the children subset. Performance using the original (non-adapted) ground truth is provided in Table S5 for the children subset and Table S6 for the full all-ages dataset.

Comparison of detection performance across validation datasets

As observed in the MiGrowD dataset (Table 3), the longitudinal method without flagging showed high sensitivity but low specificity and very low precision. Although the static method showed slightly higher sensitivity in both 2015 CCHS-Nutrition validation datasets, the longitudinal method with flagging consistently achieved higher specificity and precision, reproducing the performance pattern

Table 3 Results of detection methods on the MIcrobiota, GROWth and Diet (MiGrowD) study data across all nutrients using the longitudinally adapted ground truth.

Nutrient	Outlier detection method	Sensitivity/ recall (%)	Specificity (%)	Precision (%)	Balanced accuracy (%)
	Longitudinal with flagging	85.7	98.9	93.7	92.3
Energy intake (kcal)	Static	84.9	96.8	83.3	91.2
	Longitudinal without flagging	100.0	52.9	28.0	76.4
	Longitudinal with flagging	77.1	98.9	93.1	88.0
Carbohydrates	Static	77.0	96.8	81.8	87.0
	Longitudinal without flagging	100.0	59.7	31.2	79.8
	Longitudinal with flagging	85.7	98.4	90.9	92.0
Fiber	Static	84.3	97.3	85.7	91.5
	Longitudinal without flagging	100.0	54.9	28.9	77.5
	Longitudinal with flagging	88.5	98.9	93.9	93.7
Proteins	Static	87.1	96.8	83.7	92.7
	Longitudinal without flagging	100.0	49.7	26.7	74.9
	Longitudinal with flagging	82.8	97.9	87.8	90.3
Sugars	Static	81.9	96.8	82.8	89.8
	Longitudinal without flagging	100.0	63.3	33.3	81.7
	Longitudinal with flagging	88.2	97.9	88.2	91.8
Total fat	Static	85.7	96.8	83.7	92.7
	Longitudinal without flagging	97.1	50.3	26.4	73.7
	Longitudinal with flagging	88.5	98.9	93.9	93.7
Vitamin C	Static	85.9	96.8	83.7	92.7
	Longitudinal without flagging	100.0	60.2	31.5	80.1
	Longitudinal with flagging	91.4	98.9	94.1	95.1
Sodium	Static	90.2	96.8	84.2	94.1
	Longitudinal without flagging	100.0	52.9	28.0	76.4

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Table 4 Results of detection methods on the 2015 Canadian Community Health Survey (CCHS)-Nutrition data (all ages) across all nutrients after removing longitudinal outliers from the ground truth.

Nutrient	Outlier detection method	Sensitivity (%)	Specificity (%)	Precision (%)	Balanced accuracy (%)
	Longitudinal with flagging	89.1	98.9	65.8	94.0
Energy intake (kcal)	Static	89.7	98.4	56.6	94.1
	Longitudinal without flagging	98.8	76.6	8.6	87.7
	Longitudinal with flagging	88.8	98.8	62.4	93.8
Carbohydrates	Static	89.4	98.3	54.3	93.9
	Longitudinal without flagging	98.8	75.9	8.4	87.3
	Longitudinal with flagging	85.8	98.4	55.2	92.1
Fiber	Static	87.6	97.8	47.7	92.7
	Longitudinal without flagging	97.6	77.2	8.7	87.4
	Longitudinal with flagging	84.3	98.6	58.6	91.5
Proteins	Static	85.2	98.1	50.8	91.7
	Longitudinal without flagging	98.8	73.7	7.7	86.3
	Longitudinal with flagging	88.2	98.2	52.6	93.2
Sugars	Static	88.8	97.6	46.0	93.2
	Longitudinal without flagging	99.1	76.7	8.7	87.9
	Longitudinal with flagging	88.8	98.6	59.5	93.7
Total fat	Static	89.4	98.1	52.1	93.8
	Longitudinal without flagging	98.8	73.9	7.8	86.3
	Longitudinal with flagging	85.8	98.1	51.3	92.1
Vitamin C	Static	86.4	97.5	43.8	92.0
	Longitudinal without flagging	99.4	76.1	8.5	87.7
	Longitudinal with flagging	89.7	98.7	61.6	94.2
Sodium	Static	90.3	98.2	53.8	94.3
	Longitudinal without flagging	98.8	73.1	7.6	85.9

observed in MiGrowD. The effect of AO injection differed between the 2 2015 CCHS-Nutrition validation datasets. Although the same proportion of recalls was modified, the changes in nutrient summary statistics were smaller in the full all-ages dataset than in either the age-matched child subset or the MiGrowD dataset (Tables S3 and S4). Consistent with this observation, precision was lower in the full

Table 5 Results of detection methods on the 2015 Canadian Community Health Survey (CCHS)-Nutrition data (ages 8-12) across all nutrients after removing longitudinal outliers from the ground truth.

Nutrient	Outlier detection method	Sensitivity (%)	Specificity (%)	Precision (%)	Balanced accuracy *(%)*
	Longitudinal with flagging	86.7	98.4	90.8	92.5
Energy intake (kcal)	Static	87.1	96.4	81.7	91.8
	Longitudinal without flagging	98.8	40.8	23.2	69.8
	Longitudinal with flagging	87.3	97.8	88.1	92.6
Carbohydrates	Static	87.9	96.2	80.8	92.0
	Longitudinal without flagging	99.0	40.7	23.2	69.8
	Longitudinal with flagging	85.6	97.8	87.7	91.7
Fiber	Static	86.7	95.9	79.4	91.3
	Longitudinal without flagging	98.4	43.3	23.9	70.9
	Longitudinal with flagging	87.1	97.9	88.4	92.5
Proteins	Static	87.7	95.7	78.9	91.7
	Longitudinal without flagging	98.8	40.3	23.1	69.6
	Longitudinal with flagging	87.1	97.3	85.8	92.2
Sugars	Static	87.5	95.5	78.0	91.5
	Longitudinal without flagging	99.2	42.3	23.8	70.8
	Longitudinal with flagging	87.3	98.0	89.1	92.7
Total fat	Static	87.9	96.4	81.5	92.1
	Longitudinal without flagging	99.0	39.8	23.0	69.4
	Longitudinal with flagging	87.3	97.5	86.5	92.4
Vitamin C	Static	88.2	95.6	78.6	91.9
	Longitudinal without flagging	99.0	42.8	23.9	70.9
	Longitudinal with flagging	86.0	97.5	86.5	91.7
Sodium	Static	86.2	96.1	80.4	91.2
	Longitudinal without flagging	99.6	40.2	23.2	69.9

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all-ages dataset, ranging from 7.6% to 65.8% (Table 4), compared with 23.0% to 89.1% in the age-matched child subset (Table 5). Dataset heterogeneity was shown to reduce precision, but the longitudinal method with flagging remained the most precise approach across the 3 validation datasets.

Discussion

This work introduces a novel reproducible pipeline to detect, explore, and explain outliers in repeated 24HR data, which was evaluated using static and longitudinal AO. The hybrid framework combined transparent statistical outlier detection based on percentile- and IQR-based thresholds with explainable ML to identify interpretable dietary patterns associated with flagged observations. We also digitized and automated the NCI guidelines for outlier detection and created reference values for children's and adults' diet derived directly from Canadian dietary data, ensuring that they reflect population-specific dietary patterns.

In the MiGrowD study, approximately one-third of cleaned 24HRs contained at least 1 flagged outlier among energy intake and the 7 examined nutrients, corresponding to 75 of 226 recalls. While NCI guidance performed well in static settings, the longitudinal method with flagging improved sensitivity when repeated outliers occurred within individuals and showed more stable true negative rates across nutrients. However, methods without flagging demonstrated high sensitivity, but low precision, indicating a tendency toward over-detection ("overtossing"). These findings highlight limitations of existing approaches and reinforce the need for context-aware methods. Moreover, age emerged as a key factor, influencing both reference values and data quality, underscoring the importance of age-specific calibration. Although absolute performance varied across datasets, longitudinal flagging reduced over-detection compared with the no-flagging approach. The lower precision observed in the larger all-ages 2015 CCHS-Nutrition dataset highlights the importance of considering dataset size and heterogeneity when applying the framework to new populations.

This work has several contributions. First, it provides an automated approach to detect and evaluate outliers that maintained a strict separation between natural and AO to preserve control over their type, location, and certainty. Second, this outlier detection framework is interpretable, which is particularly important for diet data because extreme nutrient values may arise from true episodic intake, reporting error, or food-coding issues. Rule extraction therefore helps researchers examine whether a flagged value reflects a plausible dietary pattern or a potential outlier. Third, the proposed approach is tailored to longitudinal data, which, to the best of our knowledge, has not been previously presented in the context of diet. By filling this gap, our pipeline can provide new directions in longitudinal diet analysis. Rather than assuming progressive change, our method uses repeated 24HRs to identify recurring intake patterns and reduce false positives.

Consistent with the NCI guidance,[14] our framework emphasizes identification and exploration of extreme intake values rather than automatic exclusion, supporting informed review and decision-making. Our framework automates the NCI guidance, enables population-specific reference value extraction, increases efficiency and accuracy, and supports the generation of AI-ready dietary datasets for downstream analyses. Although we extracted reference values from Canadian populations, the same code can be adapted in other datasets for future implementation in different settings. In fact, the process of extracting reference values from a population dataset, as demonstrated here in a systematic manner, supports adaptation of the pipeline across populations, countries, and age groups, provided that appropriate population-specific calibration is performed.

This study also has limitations. Although the performance of the pipeline was satisfactory, we did not test datasets with more than 4 time points. The unique nature of longitudinal diet assessment in children, but also the lack of open and publicly available datasets, were barriers in testing our pipeline with longitudinal outliers over a longer period. Moreover, the NCI 5th-95th percentile guidance that was used in this work was developed using United States survey data and is intended as a triage tool rather than fixed exclusion thresholds, which may have limitations for use in other populations. To mitigate this limitation, we derived all reference thresholds from the CCHS datasets. Future work should extend validation to more diverse and longer-term datasets. Finally, the outlier thresholds used in the modified BIV approach represent a pragmatic adaptation of an existing method rather than universally accepted cutoffs for dietary intake data. Although they were selected to provide a robust approach for skewed dietary data, future studies should evaluate alternative threshold definitions and assess their impact across different dietary assessment settings.

Conclusion

In summary, this automated diet outlier detection framework enhances efficiency, standardization, and transparency. By using repeated 24HRs, the longitudinal method helps contextualize extreme values and reduce unnecessary data loss. Consistent with NCI guidance, we showed that caution should be exercised when discarding extreme nutrient values in short-term dietary recall data, because these values may reflect true intake rather than measurement error. The automated approach supports more reproducible dietary data cleaning and the generation of AI-ready datasets in nutritional epidemiology.

Ethical standards disclosure

This study was conducted according to the guidelines laid down in the Declaration of Helsinki, and all procedures involving research study participants were approved by the Hospital for Sick Children (REB numbers 1000059644 and 1000012436 for the TARGet Kids! cohort), St. Michael's Hospital-Unity Health (REB number 18-252), and the University of Toronto (REB number 00036627). Written informed consent was obtained from all subjects.

Acknowledgments

We thank Carolyn Spiegel-Feld for assisting with data collection. We thank the children, parents, research personnel, and physicians who participated in this study and in the TARGet Kids! primary care research network (www.targetkids.ca).

Members of the TARGet Kids! Collaboration: *Co-Leads*: Catherine S. Birken, Jonathon L. Maguire; *Advisory Committee*: Ronald Cohn, Eddy Lau, Andreas Laupacis, Patricia C. Parkin, Michael Salter, Peter Szatmari, Shannon Weir-Seeley; *Executive Committee*: Christopher Allen, Laura N. Anderson, Dana Arafeh, Cornelia M. Borkhoff, Mateenah Jaleel, Charles Keown-Stoneman, Dalah Mason, Jenna Pirmohamed; *Site Investigators*: Murtala Abdurrahman, Kelly Anderson, Gordon

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Arbess, Jillian Baker, Tony Barozzino, Sylvie Bergeron, Dimple Bhagat, Gary Bloch, Joey Bonifacio, Ashna Bowry, Caroline Calpin, Douglas Campbell, Sohail Cheema, Elaine Cheng, Brian Chisamore, Evelyn Constantin, Karoon Danayan, Paul Das, Mary Beth Derocher, Anh Do, Kathleen Doukas, Anne Egger, Allison Farber, Amy Freedman, Sloane Freeman, Sharon Gazeley, Charlie Guiang, Dan Ha, Curtis Handford, Laura Hanson, Leah Harrington, Sheila Jacobson, Lukasz Jagiello, Gwen Jansz, Paul Kadar, Florence Kim, Tara Kiran, Holly Knowles, Bruce Kwok, Sheila Lakhoo, Margarita Lam Antoniades, Eddy Lau, Denis Leduc, Fok-Han Leung, Alan Li, Patricia Li, Jessica Malach, Roy Male, Vashti Mascoll, Aleks Meret, Elise Mok, Rosemary Moodie, Maya Nader, Katherine Nash, Sharon Naymark, James Owen, Michael Peer, Kifi Pena, Marty Perlmutar, Navindra Persaud, Andrew Pinto, Michelle Porepa, Vikky Qi, Nasreen Ramji, Noor Ramji, Danyaal Raza, Alana Rosenthal, Katherine Rouleau, Caroline Ruderman, Janet Saunderson, Vanna Schiralli, Michael Sgro, Hafiz Shuja, Susan Shepherd, Barbara Smiltnieks, Cinntha Srikanthan, Carolyn Taylor, Stephen Treherne, Suzanne Turner, Fatima Uddin, Meta van den Heuvel, Joanne Vaughan, Thea Weisdorf, Sheila Wijayasinghe, Peter Wong, John Yaremko, Ethel Ying, Elizabeth Young, Michael Zajdman; *Research Team*: Farnaz Bazeghi, Vincent Bouchard, Marivic Bustos, Charmaine Camacho, Dharma Dalwadi, Pamela Ruth Flores, Christine Koroshegyi, Tarandeep Malhi, Ataat Malick, Michelle Mitchell, Martin Ogwuru, Rejina Rajendran, Sharon Thadani, Julia Thompson, Laurie Thompson; *Project Team*: Mary Aglipay, Imaan Bayoumi, Sarah Carsley, Katherine Cost, Karen Eny, Laura Kinlin, Jessica Omand, Shelley Vanderhout, Leigh Vanderloo; *Applied Health Research Centre*: Christopher Allen, Bryan Boodhoo, Olivia Chan, David W. H. Dai, Judith Hall, Peter Juni, Gurpreet Lakhanpal, Gerald Lebovic, Karen Pope, Audra Stitt, Kevin Thorpe; *Mount Sinai Services Laboratory*: Rita Kandel, Michelle Rodrigues, Hilde Vandenberghe.

Author contributions

P.M., C.K.S., and E.M.C. designed the study. P.M., S.S., A.A. conducted research. C.S.B., J.L.M., R.H.J.B., and E.M.C. provided essential material. P.M. and S.S. analyzed the data. P.M. wrote the first draft of the paper. All authors contributed to the interpretation of study results and contributed to, read, and approved the final manuscript.

Supplementary material

Supplementary material is available at the *American Journal of Epidemiology* online.

Conflicts of interest

Except as described above, the authors declare no conflicts of interest relevant to this article.

Funding

This research was supported by the Canadian Institutes of Health Research (FRN# PJT-195842, FRN# PJ9-190386) and the Joannah and Brian Lawson Center for Child Nutrition, Temerty Faculty of Medicine, University of Toronto. E.M.C. was awarded the Lawson Family Chair in Microbiome Nutrition Research at the Temerty Faculty of Medicine, University of Toronto. P.M. and A.A. received an Ontario Graduate Scholarship. P.M. received scholarships from the Onassis Foundation and the Peterborough K. M. Hunter Charitable Foundation. The funders had no role in study design, collection, analysis, interpretation of data, writing of the report, and they had no restrictions regarding publication.

Disclaimer

C.S.B. reported receiving research funding from the Canadian Institutes for Health Research, Heart and Stroke Foundation of Canada, Physician Services Inc, The Edwin S.H. Leong Centre for Healthy Children, University of Toronto and Hospital for Sick Children, Centre for Addiction and Mental Health, Ontario Child Health Support Unit Impact Child Health Award, and a Walmart Community Grant through the SickKids Foundation for a study on food insecurity in the inpatient hospital setting. E.M.C. has received research support from Ocean Spray Cranberries and Lallemand Health Solutions and held contracts with Inquis Clinical Research; and has received consulting fees, speaker and travel support from Danone and Lallemand Health Solutions. She held the Lawson Family Chair in Microbiome Nutrition Research at the University of Toronto. E.M.C. has received research support from the Canadian Institutes of Health Research, the Natural Sciences and Engineering Research Council of Canada, the Banting and Best Diabetes Centre, University of Toronto, and the Joannah and Brian Lawson Centre for Child Nutrition, University of Toronto. The other authors had no conflicts of interest relevant to this article to declare.

Data availability

The code used for this work is available at https://github.com/Comelli-lab/24h-recall-outlier-exploration/tree/main. The MiGrowD data are not publicly available because they contain participant-level research data. The 2015 CCHS-Nutrition data are available through the appropriate Statistics Canada/Health Canada data access procedures.

References

1. Dao MC, Subar AF, Warthon-Medina M, et al. Dietary assessment toolkits: an overview. *Public Health Nutr*. 2019;22(3):404-418. https://doi.org/10.1017/S1368980018002951
2. Tooze JA, Midthune D, Dodd KW, et al. A new statistical method for estimating the usual intake of episodically consumed foods with application to their distribution. *J Am Diet Assoc*. 2006;106(10):1575-1587. https://doi.org/10.1016/j.jada.2006.07.003
3. Subar AF, Kirkpatrick SI, Mittl B, et al. The automated self-administered 24-hour dietary recall (ASA24): a resource for researchers, clinicians and educators from the National Cancer Institute. *J Acad Nutr Diet*. 2012;112(8):1134-1137. https://doi.org/10.1016/j.jand.2012.04.016
4. National Institutes of Health NCI. Automated Self-Administered 24-Hour (ASA24®) Dietary Assessment Tool. 2024. Accessed April 23, 2024. https://epi.grants.cancer.gov/asa24/
5. Saravia L, Moliterno P, Skapino E, et al. Food diary, food frequency questionnaire, and 24-hour dietary recall. In: Betim Cazarin CB, eds. *Basic Protocols in Foods and Nutrition*. New York, NY: Humana; 2022:223-247. https://doi.org/10.1007/978-1-0716-2345-9_15
6. Barnett V, Lewis T. *Outliers in Statistical Data*. Wiley; 1994. https://doi.org/10.1002/bimj.4710370219
7. Willett W. *Nutritional Epidemiology*. Oxford University Press; 2012.

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8. Cai H, Shu X-O, Hébert JR, et al. Variation in nutrient intakes among women in Shanghai, China. *Eur J Clin Nutr*. 2004;58(12):1604-1611. https://doi.org/10.1038/sj.ejcn.1602013
9. Wassertheil-Smoller S, Davis BR, Breuer B, et al. Differences in precision of dietary estimates among different population subgroups. *Ann Epidemiol*. 1993;3(6):619-628. https://doi.org/10.1016/1047-2797(93)90085-I
10. Basiotis PP, Welsh SO, Cronin FJ, et al. Number of days of food intake records required to estimate individual and group nutrient intakes with defined confidence. *J Nutr*. 1987;117(9):1638-1641. https://doi.org/10.1093/jn/117.9.1638
11. Nelson M, Black AE, Morris JA, et al. Between-and within-subject variation in nutrient intake from infancy to old age: estimating the number of days required to rank dietary intakes with desired precision. *Am J Clin Nutr*. 1989;50(1):155-167. https://doi.org/10.1093/ajcn/50.1.155
12. Ma Y, Olendzki BC, Pagoto SL, et al. Number of 24-hour diet recalls needed to estimate energy intake. *Ann Epidemiol*. 2009;19(8):553-559. https://doi.org/10.1016/j.annepidem.2009.04.010
13. Grubbs FE. Procedures for detecting outlying observations in samples. *Technometrics*. 1969;11(1):1-21. https://doi.org/10.1080/00401706.1969.10490657
14. National Cancer Institute. Reviewing and cleaning ASA24® data. 2020. Accessed April 23, 2024. https://epi.grants.cancer.gov/asa24/resources/asa24-data-cleaning-2020.pdf
15. National Cancer Institute. Learn more about outliers. 2014. Accessed October 8, 2026. https://epi.grants.cancer.gov/dietary-assessment-primer/learn/outliers.html
16. Blázquez-García A, Conde A, Mori U, et al. A review on outlier/anomaly detection in time series data. *ACM Comput Surv (CSUR)*. 2021;54(3):1-33. https://doi.org/10.1145/3444690
17. Foorthuis R. On the nature and types of anomalies: a review of deviations in data. *Int J Data Sci Anal*. 2021;12(4):297-331. https://doi.org/10.1007/s41060-021-00265-1
18. Massara P, Asrar A, Bourdon C, et al. New approaches and technical considerations in detecting outlier measurements and trajectories in longitudinal children growth data. *BMC Med Res Methodol*. 2023;23(1):232. https://doi.org/10.1186/s12874-023-02045-w
19. Wilkinson MD, Dumontier M, Aalbersberg IJ, et al. The FAIR guiding principles for scientific data management and stewardship. *Sci Data*. 2016;3(1):160018. https://doi.org/10.1038/sdata.2016.18
20. Massara P, Spiegel-Feld C, Hamilton J, et al. Association between gut MIcrobiota, GROWth and Diet in peripubertal children from the TARGet kids! cohort (The MiGrowD) study: protocol for studying gut microbiota at a community-based primary healthcare setting. *BMJ Open*. 2022;12(5):e057989. https://doi.org/10.1136/bmjopen-2021-057989
21. Carsley S, Borkhoff CM, Maguire JL, et al. Cohort profile: the applied research group for kids (TARGet kids!). *Int J Epidemiol*. 2015;44(3):776-788. https://doi.org/10.1093/ije/dyu123
22. Health Canada. *Canadian Community Health Survey (CCHS)*. Government of Canada; 2021. Accessed April 23, 2024. https://www.canada.ca/en/health-canada/services/food-nutrition/food-nutrition-surveillance/healthnutrition-surveys/canadian-community-health-survey-cchs.html
23. Health Canada. *Reference Guide to Understanding and Using the Data: 2015 Canadian Community Health Survey-Nutrition*. Health Canada; 2017.
24. Massara P, Asrar A, Bourdon C, et al. Outlier detection in longitudinal children growth measurements. In: Onut V, Zulkernine FH, eds. *Proceedings of the 31st Annual International Conference on Computer Science and Software Engineering*. IBM Corp; 2021:220-225. https://dl.acm.org/doi/abs/10.5555/3507788.3507821
25. Hubert M, Vandervieren E. An adjusted boxplot for skewed distributions. *Comput Stat Data Anal*. 2008;52(12):5186-5201. https://doi.org/10.1016/j.csda.2007.11.008
26. Tukey JW. *Exploratory Data Analysis*. Vol. 2. Reading, MA: Addison-Wesley; 1977:131-160.
27. Breiman L, Friedman J, Olshen RA, et al. *Classification and Regression Trees*. Chapman and Hall/CRC; 2017. https://doi.org/10.1201/9781315139470
28. Therneau T, Atkinson B. Rpart: recursive partitioning and regression trees. R package version 4.1.27. *Comprehensive R Archive Network (CRAN)*. 2026; https://cran.r-project.org/web/packages/rpart/index.html
29. Giabbanelli PJ, Adams J. Identifying small groups of foods that can predict achievement of key dietary recommendations: data mining of the UK National Diet and Nutrition Survey, 2008–12. *Public Health Nutr*. 2016;19(9):1543-1551. https://doi.org/10.1017/S1368980016000185
30. Burcham S, Liu Y, Merianos AL, et al. Outliers in nutrient intake data for US adults: national health and nutrition examination survey 2017–2018. *Epidemiol Methods*. 2023;12(1):20230018. https://doi.org/10.1515/em-2023-0018
31. Trujillano J, Serviá L, Badia M, et al. Methodological review of classification trees for risk stratification: an application example in the obesity paradox. *Nutrients*. 2025;17(11):1903. https://doi.org/10.3390/nu17111903
32. van Buuren S, Groothuis-Oudshoorn K. Mice: multivariate imputation by chained equations in R. *J Stat Softw*. 2011;45(3):1-67. https://doi.org/10.18637/jss.v045.i03
33. Spratt M, Carpenter J, Sterne JA, et al. Strategies for multiple imputation in longitudinal studies. *Am J Epidemiol*. 2010;172(4):478-487. https://doi.org/10.1093/aje/kwq137

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American Journal of Epidemiology, 2026, **00**, 1–11
https://doi.org/10.1093/aje/kwag204
Novel Methodology